

FULL MARKING SCHEME

IS2200

IS Analysis and Design for Business

Winter 2023 Examination · Paper 1

Module	IS2200 — IS Analysis and Design for Business
Session	Winter 2023
Duration	1.5 hours
Structure	Section A (compulsory) · one question from Section B
Total marks	100 (50 + 50)

SECTION A — COMPULSORY

Question 1

(a) A use case diagram outlining the identified users, and how they interact with the system. Clearly **(10 marks)** label the users, use cases, and associations.

FULL MODEL ANSWER

Scope note. Question 1 is one extended scenario (Harvey Norman's inventory tracking system) developed across parts (a)–(c). Unlike some other sittings of this paper, part (c) here explicitly asks for a **Level 1 DFD** (not "Level 0"), which matches this module's own lecture terminology exactly — context diagram first, then a Level 1 decomposition into numbered internal processes. No terminology reconciliation is needed for this sitting.

The scenario names four distinct actors, each responsible for a separate cluster of system functionality:

- **Inventory Manager** — oversees inventory levels, manages stock, and ensures accurate counts.
- **Sales Associate** — processes sales and updates inventory as products are sold.
- **Supplier** — receives and confirms orders, updates stock availability, and provides product, order-history, and shipping information.
- **Data Analyst** — explores sales trends to create promotional events and generates revenue forecasting reports.

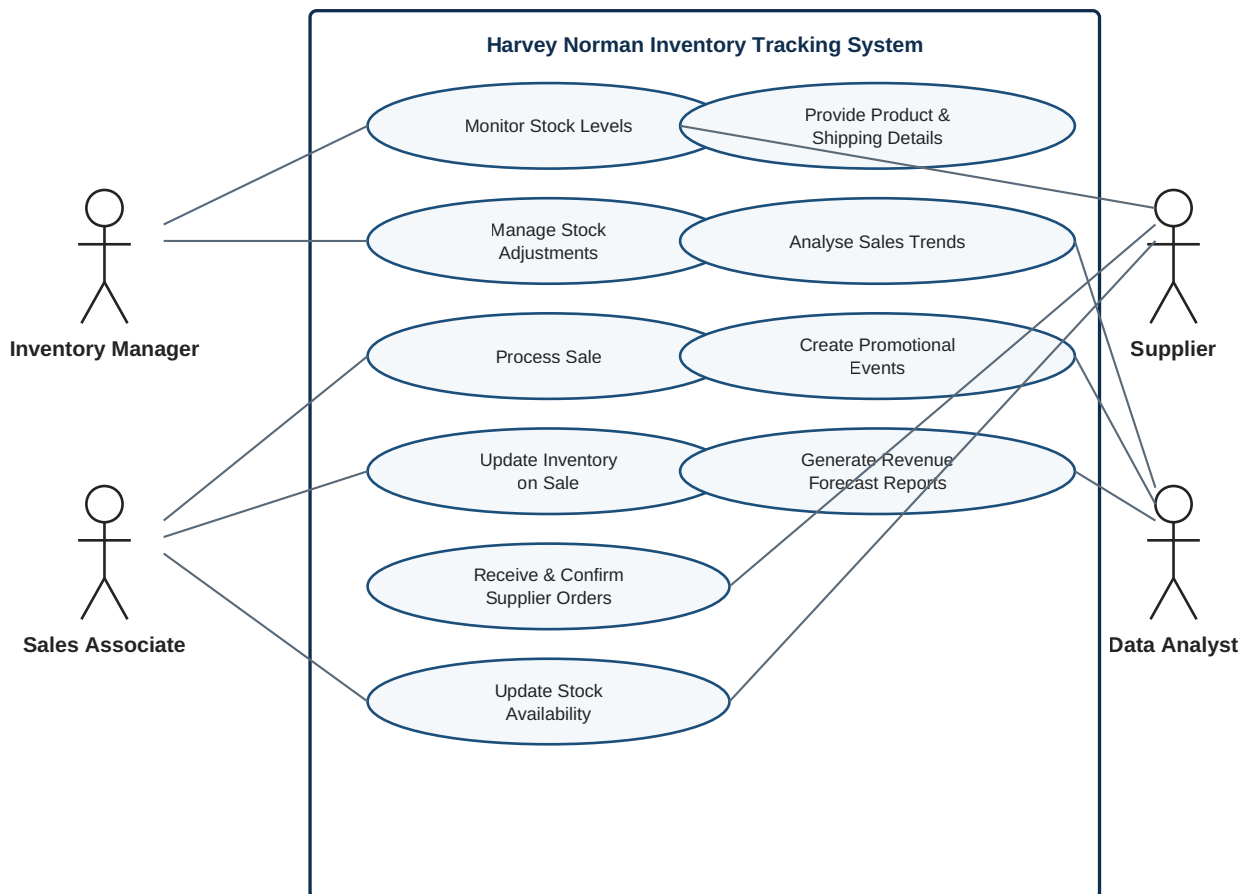


Figure 1(a). Use case diagram for the Harvey Norman inventory tracking system, showing four actors and ten associated use cases.

"Manage Stock Adjustments" is kept separate from "Monitor Stock Levels" because the scenario distinguishes oversight (monitoring) from active correction (adjusting counts) as two things the Inventory Manager does; similarly "Analyse Sales Trends," "Create Promotional Events," and "Generate Revenue Forecast Reports" are kept as three use cases rather than one, since the scenario names them as three distinct analyst outputs, not steps of a single task.

Mark Allocation — (a)

Component	Marks	Criteria for full marks
All four actors correctly identified	3	Inventory Manager, Sales Associate, Supplier, and Data Analyst all present and correctly labelled.
Use cases correctly identified and meaningfully named	4	Use cases reflect the distinct goals named in the scenario; named as verb phrases; no major function missing or wrongly merged.
Associations correctly drawn	2	Every use case connected to at least one actor; no actor left unconnected; no association to an unrelated use case.
Correct UML notation	1	System boundary box, ellipse use cases, and stick-figure actors used correctly and consistently.
Total	10	—

(b) Create a context level data flow diagram to provide an overview of the system's interactions with **(20 marks)** external entities. Label all elements appropriately.

FULL MODEL ANSWER

The context-level DFD shows the inventory tracking system as a single process, bounded by the four external entities identified in part (a) and the major labelled data flows crossing the system boundary to and from each of them.

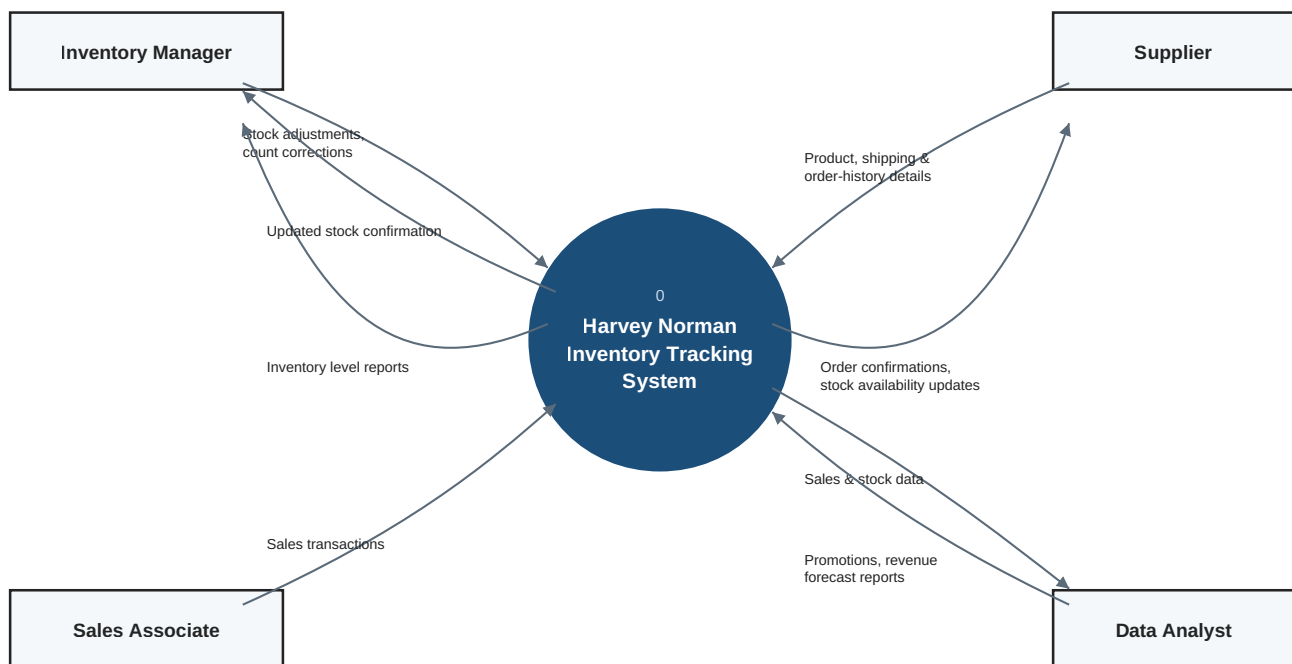


Figure 1(b). Context-level DFD: the Harvey Norman inventory tracking system as a single process (0), bounded by four external entities.

Flows are drawn directly from the scenario's own wording: *stock adjustments and count corrections* in from, and *inventory level reports* back out to, the Inventory Manager; *sales transactions* in from, and an *updated stock confirmation* back to, the Sales Associate; *product, shipping and order-history details* in from, and *order confirmations plus stock-availability updates* back to, the Supplier; and *sales and stock data* out to, with *promotional events and revenue forecast reports* back from, the Data Analyst.

Mark Allocation — (b)

Component	Marks	Criteria for full marks
Single process correctly represents the whole system	4	Exactly one process, numbered 0, labelled with the system name; no internal detail shown.
All four external entities correctly identified	8	2 marks each: Inventory Manager, Sales Associate, Supplier, and Data Analyst all present and correctly drawn as external entities.
Data flows correctly labelled and directional	6	At least one inbound and one outbound flow per entity, sensibly named from the scenario's own wording.
Correct DFD notation used throughout	2	Circle/rounded shape for the process, squares/rectangles for entities, labelled directional arrows; no data-store symbol at this level.
Total	20	—

(c) Expand on the context level data flow diagram by creating a level 1 data flow diagram.

(20 marks)

FULL MODEL ANSWER

Decomposing process 0 into its major internal processes — while keeping every external entity and flow from part (b) unchanged (the balancing rule) — gives four processes, one per actor cluster identified in the scenario:

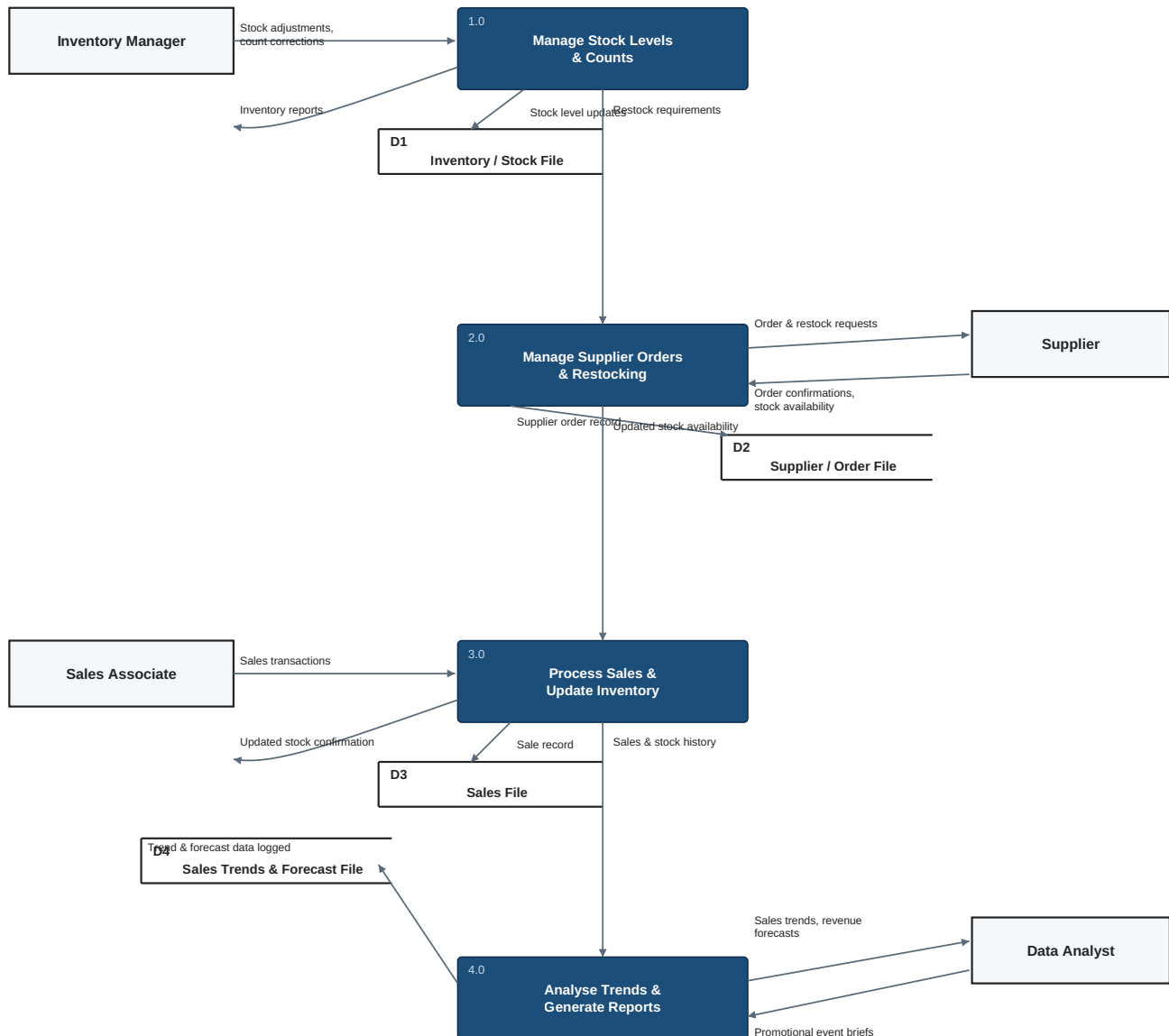


Figure 1(c). Level 1 DFD: decomposition of the Harvey Norman inventory tracking system into four numbered processes and four data stores.

Process	Covers (from the scenario)
1.0 Manage Stock Levels & Counts	Inventory Manager oversight of stock, corrections to inaccurate counts.
2.0 Manage Supplier Orders & Restocking	Supplier receives/confirms orders, updates stock availability, supplies product/shipping details, addressing the scenario's stated restocking delays.
3.0 Process Sales & Update Inventory	Sales Associate processes sales and updates inventory as items sell, replacing the old manual, error-prone entry process.
4.0 Analyse Trends & Generate Reports	Data Analyst explores sales trends, creates promotional events, and produces revenue forecasting reports.

Four data stores hold the persistent data the scenario calls out explicitly: **D1 Inventory/Stock File** (the stock counts previously kept inaccurately by manual entry), **D2 Supplier/Order File** (order and restocking records), **D3 Sales File** (transaction records as items are sold), and **D4 Sales Trends & Forecast File** (the analyst's trend data and forecasting output).

Mark Allocation — (c)

Component	Marks	Criteria for full marks
Processes correctly identified, numbered, and decomposed from the context diagram	8	4 major processes present (2 marks each), each a genuine transformation of data, consistently numbered 1.0–4.0.
Data stores correctly identified and labelled	5	D1–D4 all present, each holding data genuinely described as persisted in the scenario, correctly drawn with open-ended store notation.
Data flows accurately connect processes, stores and entities	5	Flows correctly labelled and directed; no data store connected directly to another store or external entity without an intervening process.
Consistency (balancing) with the context diagram in part (b)	1	Same four external entities appear; no external flow from part (b) is dropped or contradicted.
Correct DFD notation throughout	1	Rectangles for processes, open rectangles for stores, squares for entities, labelled directional arrows.
Total	20	—

Question 1 Total

50 marks

SECTION B · OPTION A

SECTION B — ANSWER ONE QUESTION FROM THIS SECTION

Question 2

(a) Develop five user stories that capture the features, functionality, and needs of various stakeholders interacting with the Spotify platform.

(15 marks)

FULL MODEL ANSWER

Each user story follows the standard "As a [user], I want to [achieve a goal], so that [benefit]" structure, and each targets a distinct Spotify feature named in the scenario (personalised playlists, offline downloads, collaborative playlists, podcasts) plus one further core feature (ad-free listening) to represent the range of stakeholders interacting with the platform:

#	User story	Feature represented
1	As a casual listener , I want personalised playlists generated from my listening habits, so that I can discover new music without manually searching for it.	Personalised playlists
2	As a daily commuter , I want to download my playlists for offline listening, so that I can keep listening without mobile data or a connection underground.	Offline downloads
3	As a member of a friend group , I want to create and edit a playlist together with my friends, so that everyone can contribute songs ahead of a shared event.	Collaborative playlists
4	As a podcast listener , I want to subscribe to my favourite shows and be notified of new episodes, so that I never miss new content.	Podcasts
5	As a premium subscriber , I want an ad-free listening experience, so that I can enjoy music without interruptions.	Ad-free/premium tier

Mark Allocation — (a)

Component	Marks	Criteria for full marks
Story 1 – Personalised playlists	3	Correct 'As a / I want / so that' format; goal and benefit both clearly tied to the personalised-playlist feature.
Story 2 – Offline downloads	3	Correct format; goal and benefit both clearly tied to the offline-download feature.
Story 3 – Collaborative playlists	3	Correct format; goal and benefit both clearly tied to the collaborative-playlist feature.
Story 4 – Podcasts	3	Correct format; goal and benefit both clearly tied to the podcast feature.
Story 5 – Distinct fifth stakeholder need	3	Correct format; represents a genuinely distinct stakeholder/feature from the other four stories (not a restatement).
Total	15	—

(b) Define five user behaviors for key features within the Spotify platform. These behaviors should **(15 marks)** describe how different personas interact with the system, and what actions they take to achieve their experiences. Consider how different personas may have distinct behaviors.

FULL MODEL ANSWER

Where user stories capture *what* a user wants and why, user behaviours describe *how* different personas actually act when using a given feature — the concrete pattern of interaction, not just the goal:

#	Feature	User behaviour
1	Personalised playlists	A regular listener repeatedly skips certain genres and replays others; the system logs these skip/replay patterns to refine weekly recommendation playlists.
2	Offline downloads	A commuter downloads a playlist over Wi-Fi the night before a known low-signal commute, then manually switches the app to offline mode before leaving home.
3	Collaborative playlists	A user shares a playlist link via a messaging app; multiple friends independently add tracks over the following days ahead of a shared event.
4	Podcasts	A podcast listener subscribes to a show, then listens during a fixed daily routine (e.g. a gym session or commute), frequently at 1.5× playback speed.
5	Social sharing / discovery	A user shares a "currently listening" track to their social profile immediately after discovering a song they like, prompting friends to seek it out in turn.

Each behaviour is written as an observable action-and-trigger pattern (what the persona does, and under what circumstance) rather than a goal statement, which is the key distinction between a user story and a user behaviour.

Mark Allocation — (b)

Component	Marks	Criteria for full marks
Behaviour 1 – Personalised playlists	3	Describes a concrete, observable interaction pattern (not a restated goal) tied to this feature.
Behaviour 2 – Offline downloads	3	Describes a concrete, observable interaction pattern tied to this feature.
Behaviour 3 – Collaborative playlists	3	Describes a concrete, observable interaction pattern tied to this feature.
Behaviour 4 – Podcasts	3	Describes a concrete, observable interaction pattern tied to this feature.
Behaviour 5 – Distinct fifth behaviour	3	Describes a genuinely distinct interaction pattern from the other four, not a restatement.
Total	15	—

(c) Create four personas representing different Spotify users with diverse preferences and usage patterns. Include key criteria that the development team should consider when designing and improving the platform. **(20 marks)**

FULL MODEL ANSWER

Four personas are built to be distinct along the axes that actually matter for design trade-offs: how they use the app, what they value, and where they get frustrated — each with concrete criteria the development team can design against.

Persona	Profile	Motivations & pain points	Key design criteria
The Busy Commuter (Age 29, urban professional)	Uses the app daily during a train/bus commute with patchy signal.	Wants reliable, uninterrupted listening; frustrated by playback stalling underground or playlists not downloading in time.	Robust offline mode; fast, low-friction playlist download before leaving Wi-Fi; low data usage in "online" fallback.
The Social Discoverer (Age 19, student)	Builds and shares collaborative playlists constantly; discovers most new music through friends.	Motivated by social validation and shared experience; frustrated when playlist edits from friends don't sync quickly.	Seamless real-time collaborative editing; easy sharing to social platforms; visible activity/attribution per contributor.
The Podcast Enthusiast (Age 45, working parent)	Uses the app mainly for podcasts during a commute and while exercising.	Values organisation and control over playback; frustrated by losing their place or forgetting new episodes.	Reliable episode-progress tracking, variable playback speed, and proactive new-episode notifications.
The Budget-Conscious Free User (Age 22, early-career)	Uses the free, ad-supported tier; highly price-sensitive.	Motivated by discovering music at no cost; frustrated by intrusive ads and unclear upgrade value.	Non-intrusive, well-paced ad placement; a clear, low-pressure upgrade path; data-efficient streaming-quality options.

Mark Allocation — (c)

Component	Marks	Criteria for full marks
The Busy Commuter	5	Realistic demographic profile, clear motivation/pain point, and design criteria directly addressing that pain point.
The Social Discoverer	5	Realistic demographic profile, clear motivation/pain point, and design criteria directly addressing that pain point.
The Podcast Enthusiast	5	Realistic demographic profile, clear motivation/pain point, and design criteria directly addressing that pain point.
The Budget-Conscious Free User	5	Realistic demographic profile, clear motivation/pain point, and design criteria directly addressing that pain point; the four personas together cover genuinely distinct usage patterns, not near-duplicates.
Total	20	—

Question 2 Total

50 marks

SECTION B · OPTION B

SECTION B — ANSWER ONE QUESTION FROM THIS SECTION

Question 3

(a) Calculate the expected value of the testing phase.

(20 marks)

FULL MODEL ANSWER

Expected value is the sum, over every possible outcome, of that outcome's probability multiplied by its financial value:

Outcome	Probability	Value (\$)	Probability × Value
Successful Testing	30%	+10,000	+3,000
Minor Bugs Found	20%	−2,000	−400
Moderate Bugs Found	20%	−5,000	−1,000
Major Bugs Found	15%	−15,000	−2,250
Critical Bugs Found	15%	−30,000	−4,500
Expected Value	100%	—	−\$5,150

The five probabilities sum to 100%, confirming the outcomes are exhaustive and mutually exclusive. The expected value of the testing phase is **−\$5,150**: averaged across many projects with this same risk profile, the testing phase should be expected to cost the project \$5,150 net, even though the single most likely individual outcome is Successful Testing (30%). This is the standard EV insight: the most probable single outcome and the probability-weighted average outcome are not the same figure once the higher-cost, lower-probability tail outcomes (major/critical bugs) are weighted in.

Mark Allocation — (a)

Component	Marks	Criteria for full marks
Correct formula/method applied	4	States or clearly applies $EV = \sum(\text{probability} \times \text{value})$.
Correct substitution of all five outcomes	8	Each of the 5 probability×value products correctly calculated (1.6 marks each).
Correct final arithmetic and expected value stated	6	Products correctly summed to −\$5,150, with correct sign.
Clear presentation and brief interpretation	2	Working shown in a table or equivalent structured form; sign/meaning of a negative EV briefly interpreted.
Total	20	—

(b) Populate a Stakeholder Power-Interest Grid based on the information provided in Table 1.

(20 marks)

FULL MODEL ANSWER

A Power–Interest grid plots each stakeholder on **power** (their ability to materially enable or block project decisions or resources) and **interest** (how much project outcomes affect their own goals or operations), producing four quadrants that each imply a different default engagement strategy.



Figure 3(b). Power–Interest grid for the Information Systems project, plotting the seven stakeholders from Table 1.

ID	Stakeholder	Power	Interest	Quadrant
A	Chief Information Officer	High — allocates resources, makes strategic decisions	High — direct interest in project's success	Manage Closely
B	Chief Financial Officer	Low — explicitly stated low power over output	High — budget and cost control	Keep Informed
C	End Users	Moderate — feedback can influence requirements	High — want a user-friendly system	Manage Closely
D	Tech Solutions Inc. (vendor)	High — expertise and contractual obligations	High — delivering a successful project is their core interest	Manage Closely
E	Sales Team	Low — low degree of power to submit change requests	Low — explicitly stated low interest	Monitor
F	Chief Executive Officer	High — senior executive, funding authority	High — directly responsible for funding and success	Manage Closely
G	Chief Sustainability Officer	High — power over system acceptance	Low — explicitly stated low interest in final product	Keep Satisfied

End Users (C) are the one genuinely arguable placement: Table 1 explicitly describes their power as "moderate," not high, which would put them in "Keep Informed" under a strict reading of the grid's two-value (high/low) logic. They are placed in "Manage Closely" here because their feedback directly shapes system requirements — a real, if indirect, lever over the project's design — combined with their explicitly high interest. A student placing End Users in "Keep Informed" instead, with a clear justification that their power is only moderate rather than high, should receive full credit for that stakeholder.

Mark Allocation — (b)

Component	Marks	Criteria for full marks
Grid correctly constructed	4	Both axes present and correctly labelled (Power vertical, Interest horizontal); four quadrants clearly delineated and named.
Placement of the 7 stakeholders	16	Just over 2 marks per stakeholder: correct or well-justified quadrant, with a brief reason grounded in each stakeholder's actual description in Table 1 (not just the label alone).
Total	20	—

(c) Discuss how this tool may help project managers make strategic decisions and prioritize stakeholder engagement in the context of this project.

(10 marks)

FULL MODEL ANSWER

The Power–Interest grid turns a flat stakeholder list into a small number of engagement tiers, letting a project manager allocate limited engagement effort defensibly rather than by instinct:

1. Concentrate governance effort on the "Manage Closely" cluster. The CIO, CEO, Tech Solutions Inc., and End Users all sit here. Because each can materially affect the project's success or requirements, the grid tells the PM these relationships need structured, frequent engagement — e.g. steering-committee updates for the CIO/CEO, and structured feedback sessions or usability testing for End Users — rather than ad-hoc contact.

2. Don't under-invest in "Keep Informed" stakeholders just because their formal power is low. The CFO has explicitly low power over project output but high interest in cost control. The grid's discipline here is to keep the CFO regularly informed on budget status specifically (not general project detail), both because their interest is genuine and because a surprised CFO late in a project is a common source of funding-related risk.

3. Use "Keep Satisfied" stakeholders to secure sign-off without over-communicating day-to-day detail. The Chief Sustainability Officer has high power (over system acceptance) but low interest in the detail of the final product. The grid recommends periodic, high-level updates focused specifically on what they need to sign off, rather than the same depth of communication given to high-interest stakeholders — avoiding both an unpleasant surprise at acceptance and unnecessary engagement overhead.

4. Apply minimal effort to "Monitor" stakeholders, but re-check the grid over time. The Sales Team currently has low power and low interest, so requires little proactive engagement — but if the new system later changes how sales data is captured or reported, their interest could rise sharply, and the PM should re-plot the grid at major milestones rather than treating this placement as fixed for the life of the project.

Together, this lets the PM prioritise a small number of relationships that could genuinely block or accelerate the project (Manage Closely), while still covering lower-power stakeholders with a fit-for-purpose, rather than uniform, level of communication.

Mark Allocation — (c)

Band	Marks	Descriptor
Excellent	9–10	Explains at least 3 distinct strategic uses of the grid, each explicitly grounded in this project's own stakeholders from Table 1 — not generic PM theory alone.
Good	6–8	Explains 2 strategic uses with reasonable grounding in the specific stakeholders, or 3 uses with only partial grounding.
Partial	3–5	Correctly describes what the grid shows and gives at least one generic strategic implication, with little connection back to Table 1.
Minimal	0–2	States that the grid helps prioritise stakeholders without explaining how or why, or largely restates part (b) without new analysis.

Question 3 Total

50 marks